# CV

## GENERAL INFORMATION



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|---|---|
| **Name** | Hilda Sandström |
| **Position** | Marie Skłodowska-Curie postdoctoral researcher |
| **Institution** | Technical University of Munich, Germany |
| **Email** | hilda.sandstroem@tum.de |
| **ORCID** | 0000-0001-7845-1088 |
| **Google Scholar** | scholar.google.com |



## CORE COMPETENCES

Scientific leadership · Project management · Molecular modelling & simulation · Structure prediction · Cheminformatics · Machine learning for chemistry · High-performance computing (HPC) · Student supervision & mentoring · Interdisciplinary collaboration · Scientific communication

## EXPERIENCE

**Since 10/2025** — **Marie Skłodowska-Curie postdoctoral researcher**
*Technical University of Munich, Germany*
Main project: Machine learning–based compound identification with mass spectrometry
- Develop machine learning models for mass spectrometry signal prediction and dataset similarity analysis.
- Develop protocols for simulating mass spectrometry data for atmospheric compounds.
- Coordinate interdisciplinary projects and supervise students.

**9/2024 – 9/2025** — **Visiting postdoctoral researcher**
*University of Gothenburg, Sweden*
Simulated mass spectrometry signals using machine learning models, molecular dynamics, reaction exploration and quantum chemistry.

**9/2022 – 9/2025** — **Postdoctoral researcher**
*Aalto University, Finland*
Main project: Machine learning–based compound identification with mass spectrometry
- Developed machine learning models for mass spectrometry signal prediction and dataset similarity analysis.
- Designed molecular descriptors enabling interpretable machine learning models.
- Benchmarked models and descriptors for reaction rate prediction.
- Coordinated interdisciplinary projects and supervised students.

**9/2017 – 5/2022** — **Early-stage researcher (PhD)**
*Chalmers University of Technology, Sweden*
Main project: Kinetic modeling and molecular structure prediction in polymerization reactions
- Applied steered molecular dynamics, density functional theory, umbrella sampling, and metadynamics for reaction pathway exploration and free-energy profiling.
- Predicted crystal structures of molecular co-crystals and identified plausible reaction products from kinetics/thermodynamics.
- Coordinated multi-site collaborations on crystal structure prediction and lipid conformer analysis; advised students.

## EDUCATION

**9/2017 – 5/2022** — **PhD in Chemistry (Theoretical chemistry)**
*Chalmers University of Technology, Sweden*
Award date 02/06/2022. Thesis: *Nitriles in Prebiotic Chemistry and Astrobiology*. Supervisor: Prof. Martin Rahm.

**8/2012 – 9/2017** — **MEng in Chemical engineering with engineering physics**
*Chalmers University of Technology, Sweden*
Award date 08/11/2017.

**8/2015 – 9/2017** — **MSc in Engineering physics (Nanotechnology master program, integrated)**
*Chalmers University of Technology, Sweden*
Award date 08/11/2017.

**8/2012 – 6/2015** — **BSc in Chemical engineering with engineering physics (integrated)**
*Chalmers University of Technology, Sweden*
Award date 12/06/2015.

## TEACHING

### LECTURES AND EXERCISES

**2018–2020** — **Quantum engineering**
*Chalmers University of Technology*
Computer labs · 1st year MSc Nanotechnology · 2 h/week

**2018–2021** — **Physical chemistry**
*Chalmers University of Technology*
Tutorials and experimental labs · 2nd year BSc Biotechnology · 12 h/week

**2018–2021** — **Theoretical chemistry**
*Chalmers University of Technology*
Computer labs · 3rd year BSc Chemical engineering · 4 h/week

**2017–2018** — **Chemistry and biochemistry**
*Chalmers University of Technology*
Experimental labs · 1st year BSc Chemical engineering · 8 h/week

**2014** — **Calculus**
*Chalmers University of Technology*
Exercise sessions · 1st year BSc Chemical engineering · 1 h/week

### PEDAGOGICAL TRAINING

**2/2026 – 4/2026** — **Doctoral Supervision Training**
*Joint leadership-skills training — Helmholtz, Max Planck Society, LMU Munich & TUM, Munich*
26 Feb – 30 Apr 2026. 32 hours equivalent.

**WS 2025/26** — **Supervising Students' Theses**
*TUM ProLehre, online workshop*
9 hours, Category E — Advising & Counseling

**2019** — **Teaching, learning and evaluation**
*Chalmers University of Technology*
3 ECTS

### SUPERVISION OF STUDENTS

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|---|---|
| **2024–2025** | Supervisor of MSc student — Aalto University |
| **Since 2024** | Advisor of 2 PhD students — Aalto University |
| **Since 2022** | Co-supervisor of PhD student — University of Helsinki |
| **2018–2024** | Supervised or co-supervised 22 BSc and visiting undergraduate students — Chalmers University of Technology, Aalto University |

## PROFESSIONAL DEVELOPMENT

### TRAINING

**6–7/2026** — **CSC Summer School in High-Performance Computing 2026**
*CSC – IT Center for Science, Nuuksio, Finland*
23 June – 2 July 2026. HPC programming with C++. 5 ECTS or 70 hours.

**2/2026 – 4/2026** — **Doctoral Supervision Training**
*Joint leadership-skills training — Helmholtz, Max Planck Society, LMU Munich & TUM, Munich*
26 Feb – 30 Apr 2026. 32 hours equivalent.

**3/2025 – 4/2025** — **Cooking Open Science!**
*Aalto University*
Blended course on open science principles for doctoral and postdoctoral researchers

### LANGUAGE COURSES

**10–12/2025** — **German for Researchers, A1.2**
*TUM Language Center*
Grade: 1.3 (very good) · 2 SWS

**1–3/2026** — **German for Researchers, A2.2**
*TUM Language Center*
Grade: 2.0 (good) · 2 SWS

## SKILLS AND COMPETENCES

### PROGRAMMING

Python, MATLAB, Bash — Well experienced

### MACHINE LEARNING AND CHEMINFORMATICS

Scikit-learn, TensorFlow, RDKit, OpenBabel, ASE — Experienced

### MOLECULAR DYNAMICS AND SIMULATION

CP2K, GROMACS, PLUMED — Expert · xTB, QCxMS, VMD — Experienced

### HIGH-PERFORMANCE COMPUTING (HPC)

Parallel computing, cluster resource management — Experienced

### VERSION CONTROL

Git — Experienced

### LANGUAGES

Swedish (Excellent) · English (Excellent) · Italian (Intermediate) · German (A2) · French (Basic)

## AWARDS AND HONOURS

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|---|---|
| 2025 | **Marie Skłodowska-Curie postdoctoral fellowship**<br>202,000 EUR |
| 2024–2025 | **LUMI extreme scale access resource allocation** |
| 2018–2021 | **Travel grants**<br>Nils Philblad Foundation (2021) · Karl and Annie Leon's Foundation (2018–2019) |

## ACADEMIC SERVICE

| | |
|---|---|
| 2026 | Poster judge — JpGU-AGU Joint Meeting 2026, Chiba, Japan |
| 2026 | Thesis reviewer and opponent — Universitat Autònoma de Barcelona (UAB), Barcelona, Spain |
| 2025 | Reviewer for *ACS Earth Space Chem*, *ACS Omega* and *Atmospheric Chemistry and Physics* |
| 2025 | Organizing committee — Nordic Workshop on AI for Climate Change, Sweden |
| Since 2025 | Core member, organizer and Finland representative — Climate AI Nordics Network; contributes job ads to the monthly newsletter |
| 2024 | Panelist on AI in chemistry, physics, and education — FysKemDagarna |
| 2023 | Organizer of workshop hands-on session — Shaking Up Tech, Workshop for underrepresented groups in STEM, Aalto University, Finland |
| 2023 | Session chair and organizer — ESTML, Levi, Finland |
| 2022 | Session chair — AbSciCon, USA |

## PEER-REVIEWED PUBLICATIONS

*15 peer-reviewed articles, 6 first author. Total citations: 157, h-index: 6, i10-index: 5 — Google Scholar, updated 7 July 2026*

15. Cappelletti, M., **Sandström, H.**, & Rahm, M. *ACS Central Science*, 12, 111–121 (2026). DOI: 10.1021/acscentsci.5c01497.
14. Lind, L., **Sandström, H.**, & Rinke, P. *The Journal of Chemical Physics*, 164 (2026). DOI: 10.1063/5.0308548.
13. Madan, I., Aliabadi, S. A., Huhtasaari, J., Matic, E., Hogedal, E., Kamińska, K., Nilsson, F., Stark, A., Izquierdo-Ruiz, F., **Sandström, H.**, Rahm, M. *QRB Discovery*, 6, e23 (2025). DOI: 10.1017/qrd.2025.10012. *[Supervised students and co-created workflow for testing stability of polymers.]*
12. Brean, J., Bortolussi, F., Rowell, A., Beddows, D. C. S., Weinhold, K., Mettke, P., Merkel, M., Kumar, A., Barua, S., Iyer, S., Karppinen, A., **Sandström, H.**, Rinke, P., et al. *ACS ES&T Air*, 2, 1704–1713 (2025). DOI: 10.1021/acsestair.5c00119. *[Supervised PhD student F. Bortolussi in developing and evaluating the machine learning model and workflow.]*
11. Izquierdo-Ruiz, F., Cable, M. L., Hodyss, R., Vu, T. H., **Sandström, H.**, Lobato, A., & Rahm, M. *Proc. Natl. Acad. Sci. U.S.A.*, 122, e2507522122 (2025). DOI: 10.1073/pnas.2507522122. *[Developed and tested crystal structure prediction program workflow for molecular cocrystals.]*
10. Valiev, R. R., Nasibullin, R. T., **Sandström, H.**, Rinke, P., Puolamäki, K., & Kurten, T. *Physical Chemistry Chemical Physics*, 27, 14804–14814 (2025). DOI: 10.1039/D5CP01101A. *[Co-advisor for ML workflow; developed MBTR model.]*
9. Bortolussi, F., **Sandström, H.**, Partovi, F., Mikkilä, J., Rinke, P., & Rissanen, M. *Atmospheric Chemistry and Physics*, 25, 685–704 (2025). DOI: 10.5194/acp-25-685-2025. *[Co-designed study, advised, and contributed to programming and model testing.]*
8. Malaska, M. J., **Sandström, H.**, Hofmann, A. E., Hodyss, R., Rensmo, L., van der Meulen, M., Rahm, M., Cable, M. L., & Lunine, J. I. *Astrobiology*, 25, 367–389 (2025). DOI: 10.1089/ast.2024.0125. *[Performed geometry optimizations, conformer search and student supervision.]*
7. **Sandström, H.**, & Rinke, P. *Geoscientific Model Development*, 18, 2701–2724 (2025). DOI: 10.5194/gmd-18-2701-2025.
6. **Sandström, H.**, Rissanen, M., Rousu, J., & Rinke, P. *Advanced Science*, 11, 2306235 (2024). DOI: 10.1002/advs.202306235.
5. **Sandström, H.**, Izquierdo-Ruiz, F., Cappelletti, M., Dogan, R., Sharma, S., Bailey, C., & Rahm, M. *ACS Earth and Space Chemistry*, 8, 1272–1280 (2024). DOI: 10.1021/acsearthspacechem.4c00088.
4. **Sandström, H.**, & Rahm, M. *The Journal of Physical Chemistry A*, 127, 4503–4510 (2023). DOI: 10.1021/acs.jpca.3c01504.
3. **Sandström, H.**, & Rahm, M. *ACS Earth and Space Chemistry*, 5, 2152–2159 (2021). DOI: 10.1021/acsearthspacechem.1c00195.
2. **Sandström, H.**, & Rahm, M. *Science Advances*, 6, eaax0272 (2020). DOI: 10.1126/sciadv.aax0272.
1. Lindblom, A., Sriram, K. K., Müller, V., Öz, R., **Sandström, H.**, Åhrén, C., Westerlund, F., & Karami, N. *Diagnostic Microbiology and Infectious Disease*, 93, 380–385 (2019). DOI: 10.1016/j.diagmicrobio.2018.10.014. *[Performed fluorescence microscopy assays where I stained, trapped, and photographed plasmids in nanochannels.]*

## TALKS

### INVITED SEMINARS AND KEYNOTES

| | |
|---|---|
| 2026 | **Data-driven mass spectrometry for atmospheric compound identification**<br>*University of Gothenburg, Sweden* |
| 2026 | **Data-driven compound identification with atmospheric mass spectrometry**<br>*Network on Mathematical Data Science for Materials Science, Workshop on the Interface of Mathematics and Machine Learning for Applications in Materials Science — University of Glasgow, UK* |
| 2025 | **CLOUDMAP: advanced identification of atmospheric compounds**<br>*Atmospheric day, Sweden — **Keynote*** |
| 2025 | **Machine learning for atmospheric mass spectrometry**<br>*Nordic Workshop on AI for Climate Change, Sweden* |
| 2024 | **AI in chemistry: solving experimental challenges with artificial intelligence**<br>*FysKemDagarna (Physics and Chemistry Days), Sweden* |

### CONTRIBUTED TALKS

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|---|---|
| 2026 | **Towards atmospheric compound identification using simulated electron ionization mass spectra**<br>*JpGU-AGU Joint Meeting 2026, Chiba, Japan* |
| 2026 | **Towards atmospheric compound identification using simulated electron ionization mass spectra**<br>*Chemical Compounds Space Conference (CCSC 2026), Munich, Germany* |
| 2023 | **Characterizing atmospheric molecules for machine learning**<br>*International Aerosol Modeling Algorithms Conference, USA* |
| 2023 | **Characterizing atmospheric molecules for machine learning**<br>*European Aerosol Conference, Spain* |
| 2023 | **Characterizing atmospheric molecules for machine learning**<br>*Physics Days, Finland* |
| 2022 | **Untangling hydrogen cyanide polymerization using quantum chemistry**<br>*AbSciCon, USA* |